

Lecture 3-1: chemical formulas, equations, and reaction yields

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Classroom: Teaching Center 401

Wednesday/Friday: 10:15-11:00; 11:10-11:55

2025/09/26

Quiz Friday!

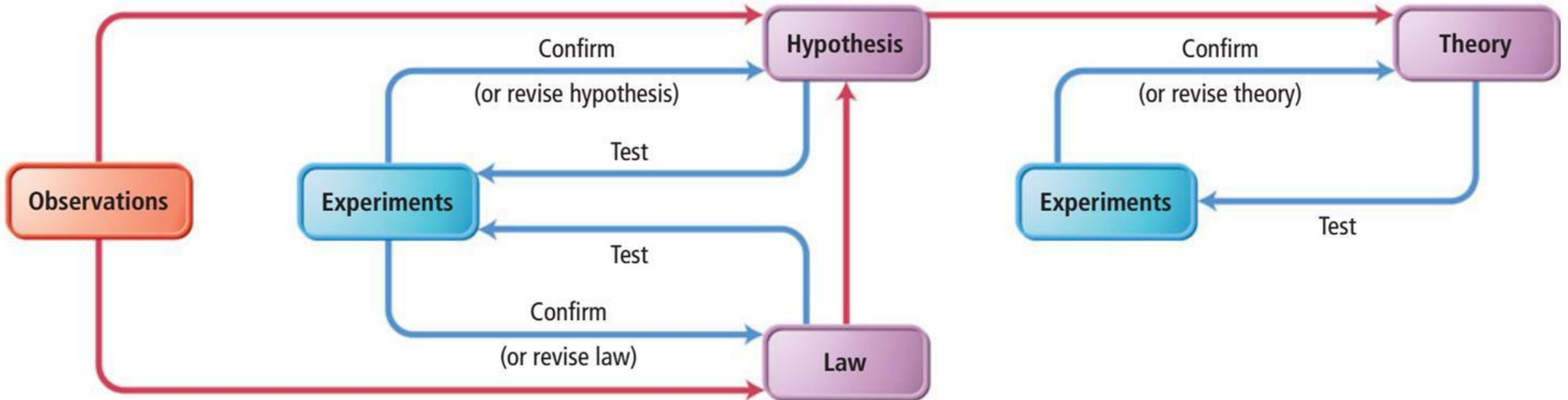
Major points of last's lecture

- **The atom in modern chemistry**

1. The nature of modern chemistry
2. Elements: the building blocks of matter
3. Laws of chemical combination
4. The physical structure of atoms
5. Mass spectrometer and isotopes
6. The mole

Scientific Method

Hypothesis (假设) vs. scientific law (科学定律)



Elements: the building blocks of matter

element *noun* (PART)

B2 [C]

a part of something

部分; 部件; 要素

- *List the elements that make up a perfect dinner party.*
列举出尽善尽美的晚宴所有的构成要素。
- *The movie had all the elements of a good thriller.*
这部电影具备了一部精彩的惊险片的一切要素。

element *noun* (SIMPLE SUBSTANCE)

B2 [C]

a simple substance that cannot be reduced to smaller chemical parts

元素

- *Aluminium is an element.*
铝是一种元素。

element *noun* (EARTH, AIR, ETC.)

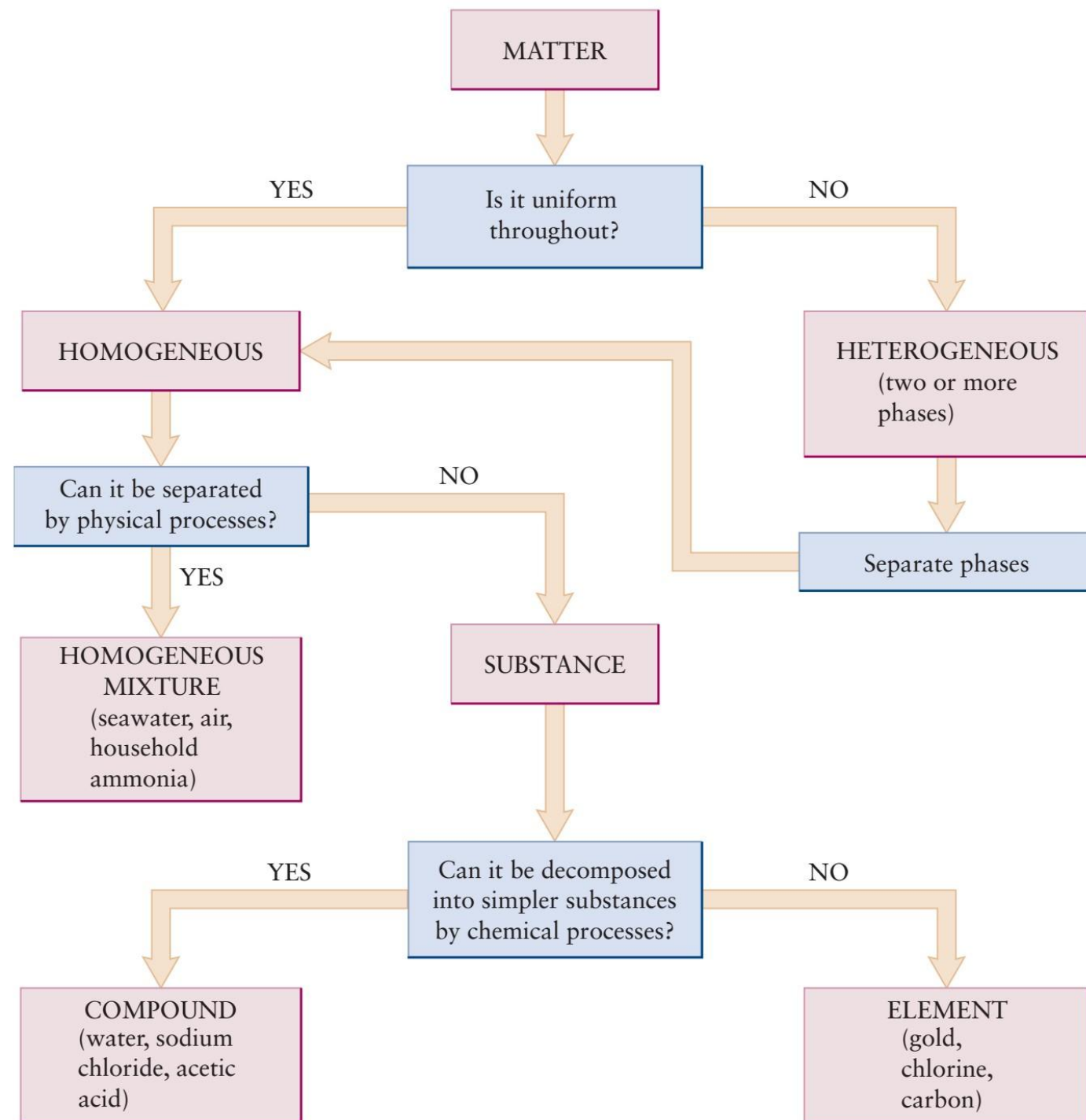
[C]

earth, air, fire, and water from which people in the past believed everything else was made

四大元素 (旧时人们认为构成一切事物的土、风、火、水)

- **Elements (单质)** are substances that **cannot be decomposed into two or more simpler substances by ordinary physical or chemical means**. The word ordinary excludes the processes of radioactive decay, whether natural or artificial, and high-energy nuclear reactions that do transform elements into one another.

- **Chemical elements (化学元素)**



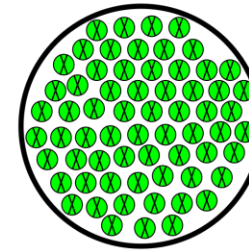
Element (单质): substance that cannot be decomposed into two or more simpler substance by ordinary physical or chemical means.

Compound (化合物) contains two or more chemical elements

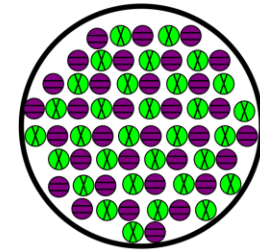
Q1: 氧气?

Q2: 水?

Pure Substances



Element



Compound

Elements are building blocks of matter.

Laws of chemical combinations

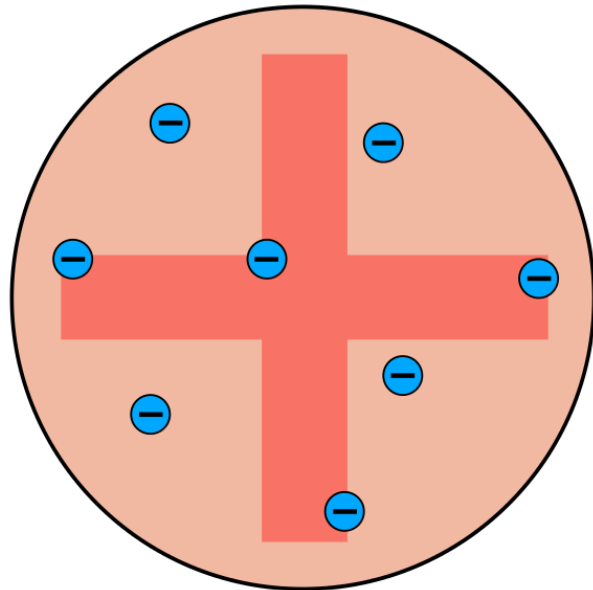
- The ratios of the masses of compounds react to form other compounds is fixed. As the smallest indivisible units of the compounds (atoms) combined to form the smallest indivisibles units of the compounds (moleculars).

1. Law of Conservation of Mass
2. Law of Definite Proportions
3. Law of Multiple Proportions
4. Law of Combining Volumes
5. Avogadro's Law

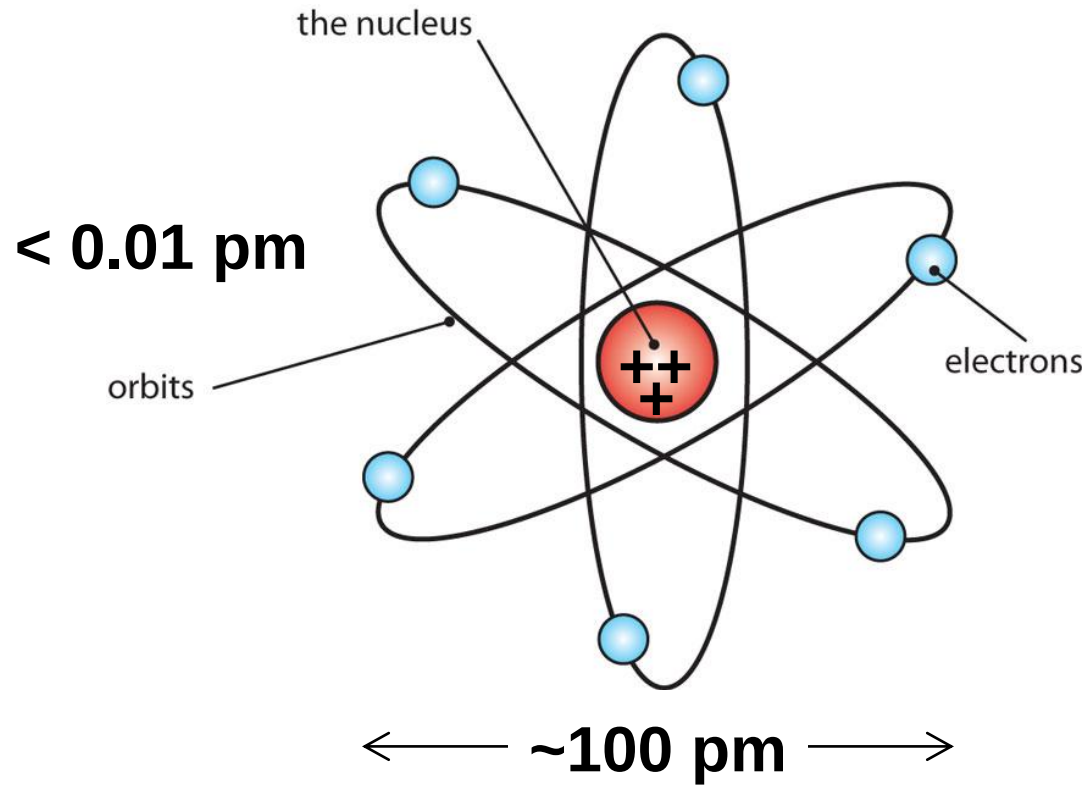
The Plum Pudding Model

J. J. Thomson

- 1897 Discovered electron
- 1904 Proposed the pudding model



The Rutherford Model



The Au nucleus has **79** positive charges
but weighs **197** mass units?

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
															Pnictogens	Chalcogens	Halogens	
1	1 H Hydrogen 1.008	Atomic Symbol Name Weight		C Solid	Metals Alkali metalsAlkaline earth metalsLanthanoidsActinoidsTransition metalsPost-transition metalsMetalloids Nonmetals Reactive nonmetalsNoble gases													2 He Helium 4.0026
2	3 Li Lithium 6.94	4 Be Beryllium 9.0122		Hg Liquid									5 B Boron 10.81	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
3	11 Na Sodium 22.990	12 Mg Magnesium 24.305		H Gas									13 Al Aluminium 26.982	14 Si Silicon 28.085	15 P Phosphorus 30.974	16 S Sulfur 32.06	17 Cl Chlorine 35.45	18 Ar Argon 39.948
4	19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.630	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.798
5	37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium (98)	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.91	46 Pd Palladium 106.42	47 Ag Silver 107.87	48 Cd Cadmium 112.41	49 In Indium 114.82	50 Sn Tin 118.71	51 Sb Antimony 121.76	52 Te Tellurium 127.60	53 I Iodine 126.90	54 Xe Xenon 131.29
6	55 Cs Caesium 132.91	56 Ba Barium 137.33	57–71 89–103	72 Hf Hafnium 178.49	73 Ta Tantalum 180.95	74 W Tungsten 183.84	75 Re Rhenium 186.21	76 Os Osmium 190.23	77 Ir Iridium 192.22	78 Pt Platinum 195.08	79 Au Gold 196.97	80 Hg Mercury 200.59	81 Tl Thallium 204.38	82 Pb Lead 207.2	83 Bi Bismuth 208.98	84 Po Polonium (209)	85 At Astatine (210)	86 Rn Radon (222)
7	87 Fr Francium (223)	88 Ra Radium (226)		104 Rf Rutherfordium (267)	105 Db Dubnium (268)	106 Sg Seaborgium (269)	107 Bh Bohrium (270)	108 Hs Hassium (277)	109 Mt Meitnerium (278)	110 Ds Darmstadtium (281)	111 Rg Roentgenium (282)	112 Cn Copernicium (285)	113 Nh Nihonium (286)	114 Fl Flerovium (289)	115 Mc Moscovium (290)	116 Lv Livermorium (293)	117 Ts Tennessine (294)	118 Og Oganesson (294)

For elements with no stable isotopes, the mass number of the isotope with the longest half-life is in parentheses.

57 La Lanthanum 138.91	58 Ce Cerium 140.12	59 Pr Praseodymium 140.91	60 Nd Neodymium 144.24	61 Pm Promethium (145)	62 Sm Samarium 150.36	63 Eu Europium 151.96	64 Gd Gadolinium 157.25	65 Tb Terbium 158.93	66 Dy Dysprosium 162.50	67 Ho Holmium 164.93	68 Er Erbium 167.26	69 Tm Thulium 168.93	70 Yb Ytterbium 173.05	71 Lu Lutetium 174.97
89 Ac Actinium (227)	90 Th Thorium 232.04	91 Pa Protactinium 231.04	92 U Uranium 238.03	93 Np Neptunium (237)	94 Pu Plutonium (244)	95 Am Americium (243)	96 Cm Curium (247)	97 Bk Berkelium (247)	98 Cf Californium (251)	99 Es Einsteinium (252)	100 Fm Fermium (257)	101 Md Mendelevium (258)	102 No Nobelium (259)	103 Lr Lawrencium (266)

Summary

Mass number

$$A = p + n$$

Atomic number
(charge number)

$$Z = p$$



Anode

阳极

Anion

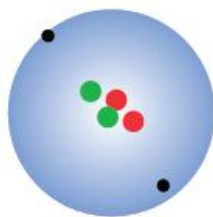
阴离子/负离子

Cathode

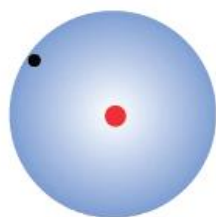
阴极

Cation

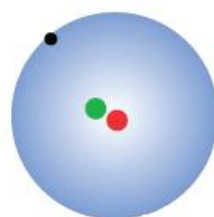
阳离子/正离子



Helium, ${}_2^4\text{He}$



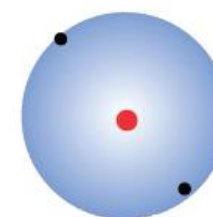
Protium, ${}_1^1\text{H}$



Deuterium, ${}_1^2\text{H}$



Hydrogen ion
(proton), H^+



Hydride, H^-

Cation

Anion

Hydrogen, ${}_1\text{H}$

Ion

Generalized hydrogen element 元素

The Mole: Weighing and Counting Molecules

- 19th century chemists concluded that oxygen atoms weigh 16 times as much as hydrogen atoms.
- In the early 20th century, relative atomic and molecular masses were determined much more accurately through the work of J. J. Thomson, F. W. Aston, and others using mass spectrometry.
- These relative masses on the atomic scale must be related to absolute masses on the gram scale by a conversion factor.

How to weigh out a sample containing **exactly the number of atoms or molecules needed for a particular chemical reaction?**

微观原子质量与 vs. 观物质质量的关系

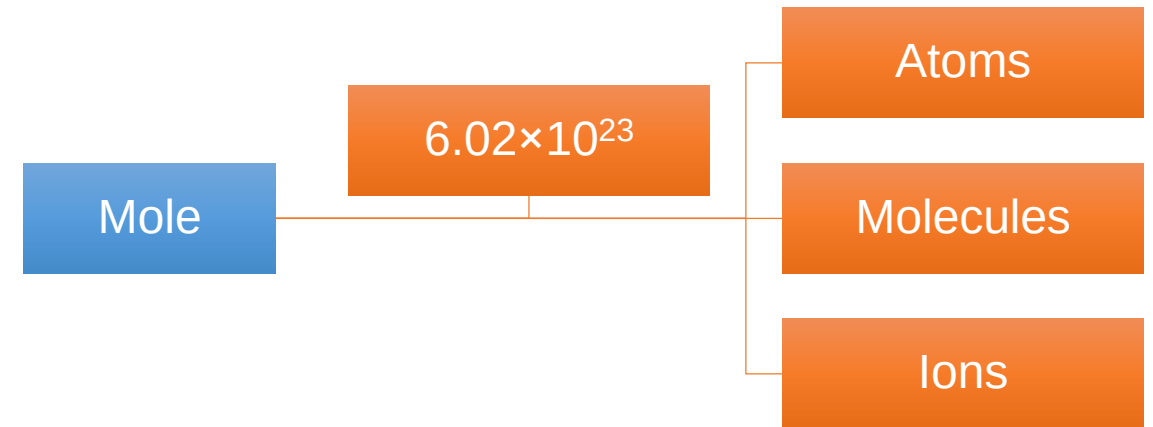
The Mole: Weighing and Counting Molecules

- **Avogadro's number (阿伏伽德罗数 N_A):** The number of atoms in exactly 12g of ^{12}C .

$N_A = 6.02214120 \times 10^{23}$ (principle of modern chemistry, 7th edition)

$$N \equiv \frac{12 \text{ g}}{m_a(^{12}\text{C})} = \frac{12 \text{ g}}{1.99 \times 10^{-23} \text{ g}} = 6.02 \times 10^{23}$$

- **The mole (摩尔数):** One mole of a substance is the amount that contains Avogadro's number of atoms, molecules, or other entities



1 mol of ^{12}C contains N_A ^{12}C atoms, 1 mol of water contains N_A water molecules.

- **The molar Mass (摩尔质量):** one mole of that a substance in grams. **Molar mass: M (g/mol)**

$$A_r(\text{Ag}) = 107.9 \quad \rightarrow \quad M(\text{Ag}) = 107.9 \text{ g mol}^{-1}.$$

$$M_r(\text{H}_2\text{O}) = 18.0 \quad \rightarrow \quad M(\text{H}_2\text{O}) = 18.0 \text{ g mol}^{-1}.$$

Density and Molecular Size

Molar volume (摩尔体积): the volume occupied by one mole of a substance:

$$V_{\text{m}} = \frac{\text{molar mass (g mol}^{-1}\text{)}}{\text{density (g cm}^{-3}\text{)}} = \text{molar volume (cm}^3 \text{ mol}^{-1}\text{)}$$

For example, near 0°C, ice has a density of 0.92 g cm⁻³; thus, the molar volume of solid water under these conditions is

$$V_{\text{m}} = \frac{18.0 \text{ g mol}^{-1}}{0.92 \text{ g cm}^{-1}} = 20 \text{ cm}^3 \text{ mol}^{-1}$$

Major points of today's lecture

Chemical formulas, equations, and reaction yields

1. Empirical and molecular formulas
2. Chemical formula and percentage composition
3. Writing balanced chemical equations
4. Mass relationships in chemical reactions
5. Limiting reactant and percentage yield

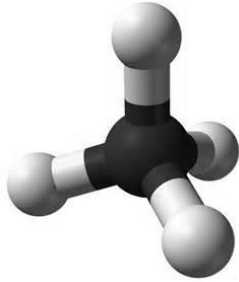
Empirical formula and the molecular formula

Molecular formula (分子式):

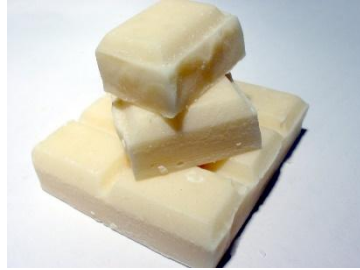
specifies the number of atoms of each element in one molecule of the substance.



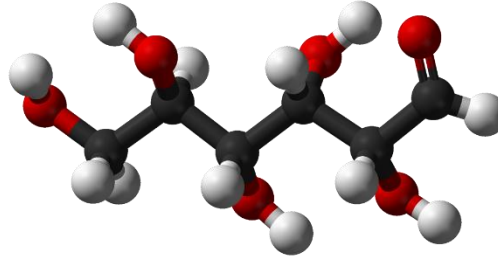
Methane



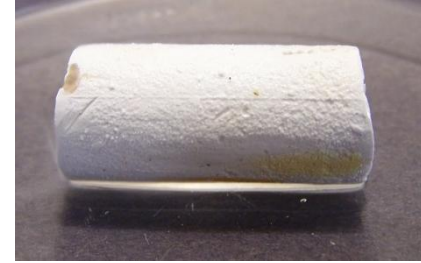
CH_4



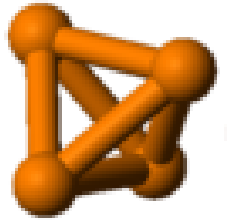
Glucose



$\text{C}_6\text{H}_{12}\text{O}_6$



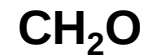
White phosphorus



P_4

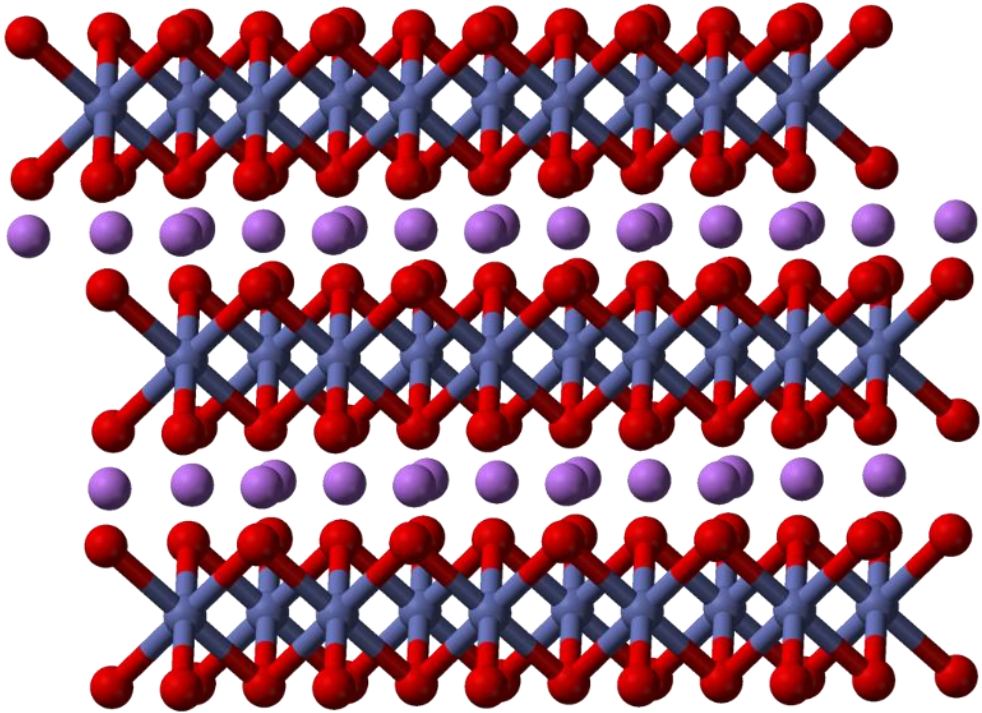
Empirical formula (经验式):

the simplest formula that give the correct relative numbers of atoms of each kind in a compound.

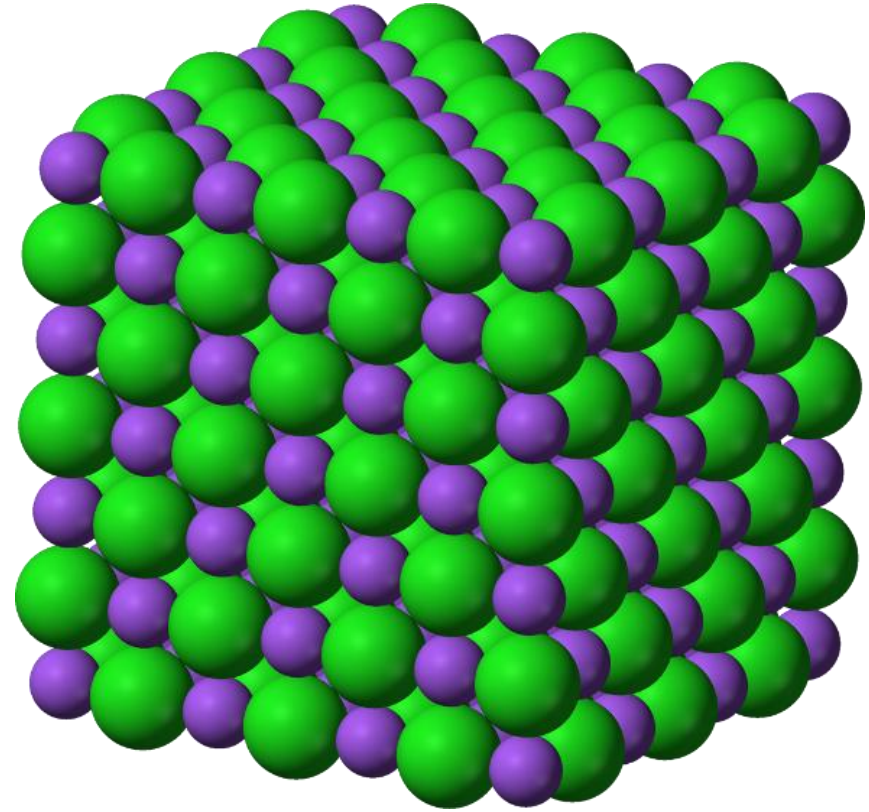


Formula unit

Formula unit (化学式单位): many solids can be represented only by their formula units because it is not possible to identify a molecular unit in a unique way.



LiCoO₂



NaCl

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Empirical formula and percentage composition

- The empirical formula for ethylene (molecular formula C_2H_4) is CH_2 .
- Its composition by mass is calculated from the masses of carbon and hydrogen in 1 mol of CH_2 formula units.

$$\text{Mass of C} = 1 \text{ mol C} \times (12.011 \text{ g mol}^{-1}) = 12.011 \text{ g}$$

$$\text{Mass of H} = 2 \text{ mol H} \times (1.00794 \text{ g mol}^{-1}) = 2.0159 \text{ g}$$

- Mass percentage of C: 85.628%
- Mass percentage of H: 14.372%

Similarly, if we know the mass percentage, we can calculate the empirical formula.

A 60.00-g sample of a dry-cleaning fluid was analyzed and found to contain 10.80 g carbon, 1.36 g hydrogen, and 47.84 g chlorine. Determine the empirical formula of the compound using a table of atomic masses.

Solution

The amounts of each element in the sample are

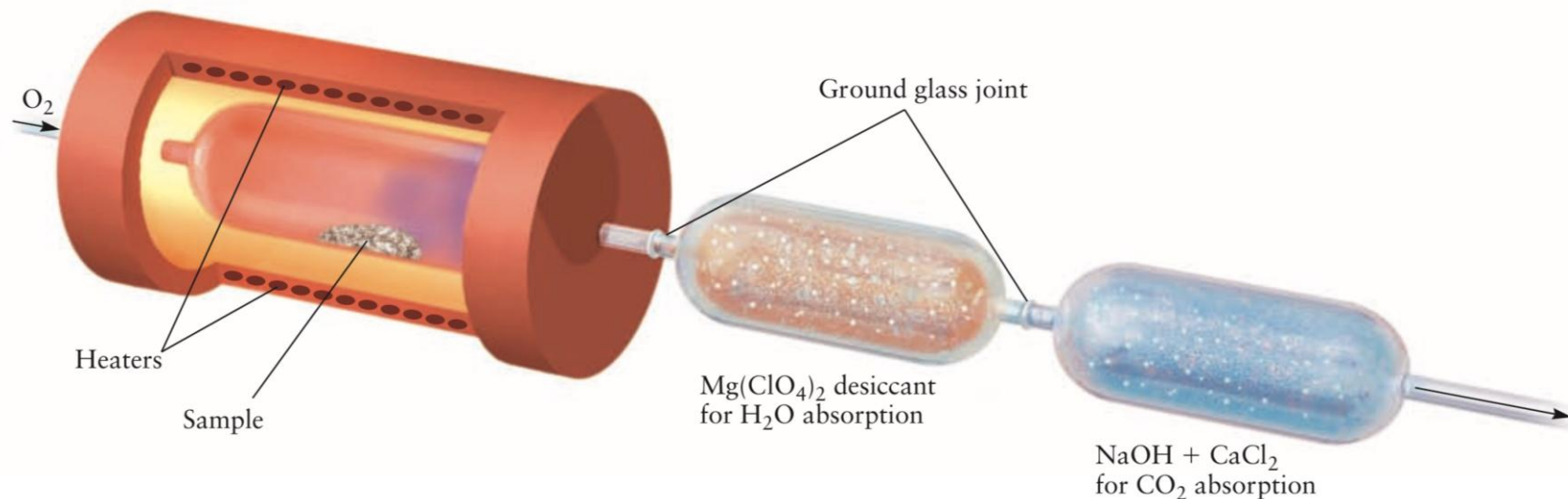
$$\text{carbon: } \frac{10.80 \text{ g C}}{12.011 \text{ g mol}^{-1}} = 0.8992 \text{ mol C}$$

$$\text{hydrogen: } \frac{1.36 \text{ g H}}{1.008 \text{ g mol}^{-1}} = 1.35 \text{ mol H}$$

$$\text{chlorine: } \frac{47.84 \text{ g Cl}}{35.453 \text{ g mol}^{-1}} = 1.349 \text{ mol Cl}$$

The ratio of the amount of carbon to that of chlorine (or hydrogen) is $0.8992:1.349 = 0.6666$, which is close to $2:3$. The numbers of moles form the ratio $2:3:3$; therefore, the empirical formula is $\text{C}_2\text{H}_3\text{Cl}_3$. Additional measurements would be necessary to find the actual molecular mass and the correct *molecular* formula from among $\text{C}_2\text{H}_3\text{Cl}_3$, $\text{C}_4\text{H}_6\text{Cl}_6$, or any higher multiples $(\text{C}_2\text{H}_3\text{Cl}_3)_n$.

Empirical Formula Determined from Elemental Analysis by Combustion



- A hydrocarbon is compound that contains only carbon and hydrogen. Its empirical formula can be determined by using the combustion train.
- In this device, a known mass of the hydrocarbon is burned completely in oxygen, yielding carbon dioxide and water.

A certain compound, used as a welding fuel, contains only carbon and hydrogen. Burning a small sample of this fuel completely in oxygen produces 3.38 g CO₂, 0.692 g water, and no other products. What is the empirical formula of the compound?

Solution

We first compute the amounts of CO₂ and H₂O. Because all the carbon has been converted to CO₂ and all the hydrogen to water, the amounts of C and H in the unburned gas can be determined:

$$\text{mol of C} = \text{mol of CO}_2 = \frac{3.38 \text{ g}}{44.01 \text{ g mol}^{-1}} = 0.0768 \text{ mol}$$

$$\text{mol of H} = 2(\text{mol of H}_2\text{O}) = \left(\frac{0.692 \text{ g}}{18.02 \text{ g mol}^{-1}} \right) = 0.0768 \text{ mol}$$

Because each water molecule contains two hydrogen atoms, it is necessary to multiply the number of moles of water by 2 to find the number of moles of hydrogen atoms. Having found that the compound contains equal numbers of moles of carbon and hydrogen, we have determined that its empirical formula is CH. Its molecular formula may be CH, C₂H₂, C₃H₃, and so on.

Connection between empirical and molecular formulas

- The molecular formula is some whole-number multiple of the empirical formula.
- From Avogadro's hypothesis (same number of molecules of two gases in same volume, temperature and pressure), the ratio of molar masses of two gaseous compounds is the same as the ratio of their densities.

乙炔的摩尔质量

乙炔的密度

氧气的摩尔质量

$$\text{Molar mass of welding gas} = \frac{1.06 \text{ g L}^{-1}}{1.31 \text{ g L}^{-1}} (32.0 \text{ g mol}^{-1}) = 25.9 \text{ g mol}^{-1}$$

氧气的密度

乙炔= C_2H_2

Major points of today's lecture

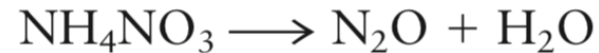
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Writing balanced chemical equations

Key: the **conservation of matter** in a chemical change is represented in a balanced chemical equation.

Stoichiometry (化学计量学): the study of the relationships between the numbers of reactant and product molecules.



Steps:

- Assign 1 as the coefficient of one species. The best choice is the most complicated species; that is, the species with the largest number of elements.
- Identify, in sequence, elements that appear in only one chemical species, the coefficient of which has not yet been determined. Choose that coefficient to balance the number of moles of atoms of that element. Continue until all coefficients have been identified.
- If desired, multiply the whole equation by the smallest integer that will eliminate any fractions.

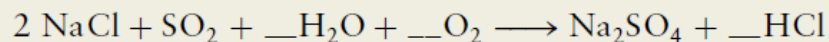
Hargreaves process is an industrial procedure for making sodium sulfate (Na_2SO_4) for use in papermaking. The starting materials are sodium chloride (NaCl), sulfur dioxide (SO_2), water, and oxygen. Hydrogen chloride (HCl) is generated as a by-product. Write a balanced chemical equation for this process.

Solution

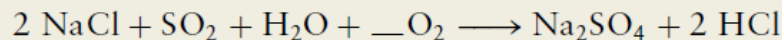
The unbalanced equation is



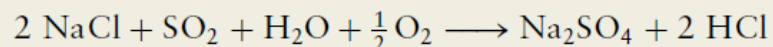
Begin by assigning a coefficient of 1 to Na_2SO_4 because it is the most complex species, composed of 3 different elements. There are 2 mol of sodium atoms on the right; therefore, the coefficient for NaCl must be 2. Following the same argument, the coefficient for SO_2 must be 1 to balance the 1 mol of sulfur on the right. This gives



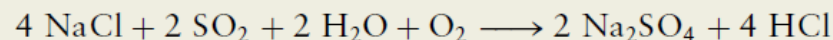
Next, we note that there are 2 mol of Cl atoms on the left (reactant) side; therefore, the coefficient for HCl must be 2. Hydrogen is the next element to balance, with 2 mol on the right side, and therefore a coefficient of 1 for the H_2O :



Finally, the oxygen atoms must be balanced. There are 4 mol of oxygen atoms on the right side, but there are 2 mol from SO_2 and 1 mol from H_2O on the left side; therefore, 1 mol of oxygen *atoms* must come from O_2 . Therefore, the coefficient for O_2 is $\frac{1}{2}$:



Multiplying all coefficients in the equation by 2 gives



In balancing this equation, oxygen was considered last because it appears in several places on the left side of the equation.

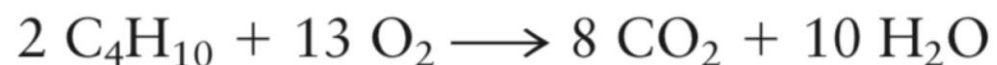
Major points of today's lecture

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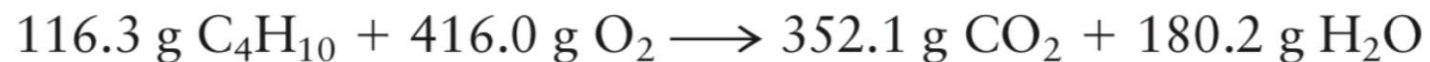
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Mass relationships in chemical reactions

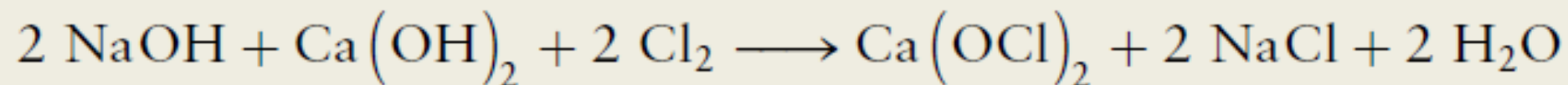
mass $\xleftrightarrow{\text{molar mass}}$ *moles* $\xleftrightarrow{N_A}$ *number of molecules*



2 molecules of C_4H_{10} + 13 molecules of O_2 $\longrightarrow$
8 molecules of CO_2 + 10 molecules of H_2O



Calcium hypochlorite, $\text{Ca}(\text{OCl})_2$, is used as a bleaching agent. It is produced from sodium hydroxide, calcium hydroxide, and chlorine according to the following overall equation:



How many grams of chlorine and sodium hydroxide react with 1067 g $\text{Ca}(\text{OH})_2$, and how many grams of calcium hypochlorite are produced?

Solution

The amount of $\text{Ca}(\text{OH})_2$ consumed is

$$\frac{1067 \text{ g Ca}(\text{OH})_2}{74.09 \text{ g mol}^{-1}} = 14.40 \text{ mol Ca}(\text{OH})_2$$

where the molar mass of $\text{Ca}(\text{OH})_2$ has been obtained from the molar masses of calcium, oxygen, and hydrogen as

$$40.08 + 2(15.999) + 2(1.0079) = 74.09 \text{ g mol}^{-1}$$

According to the balanced equation, 1 mol $\text{Ca}(\text{OH})_2$ reacts with 2 mol NaOH and 2 mol Cl_2 to produce 1 mol $\text{Ca}(\text{OCl})_2$. If 14.40 mol of $\text{Ca}(\text{OH})_2$ reacts completely, then

$$\text{mol NaOH} = 14.40 \text{ mol Ca}(\text{OH})_2 \left(\frac{2 \text{ mol NaOH}}{1 \text{ mol Ca}(\text{OH})_2} \right)$$

$$= 28.80 \text{ mol NaOH}$$

$$\text{mol Cl}_2 = 14.40 \text{ mol Ca}(\text{OH})_2 \left(\frac{2 \text{ mol Cl}_2}{1 \text{ mol Ca}(\text{OH})_2} \right)$$

$$= 28.80 \text{ mol Cl}_2$$

$$\text{mol Ca}(\text{OCl})_2 = 14.40 \text{ mol Ca}(\text{OH})_2 \left(\frac{1 \text{ mol Ca}(\text{OCl})_2}{1 \text{ mol Ca}(\text{OH})_2} \right)$$

$$= 14.40 \text{ mol Ca}(\text{OCl})_2$$

From the number of moles and molar masses of reactants and products, the following desired masses are found:

$$\text{Mass NaOH reacting} = (28.80 \text{ mol})(40.00 \text{ g mol}^{-1}) = 1152 \text{ g}$$

$$\text{Mass Cl}_2 \text{ reacting} = (28.80 \text{ mol})(70.91 \text{ g mol}^{-1}) = 2042 \text{ g}$$

$$\text{Mass Ca}(\text{OCl})_2 \text{ produced} = (14.40 \text{ mol})(142.98 \text{ g mol}^{-1}) = 2059 \text{ g}$$

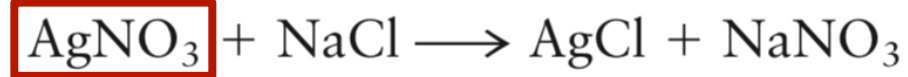
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Limiting reactant and percentage yield

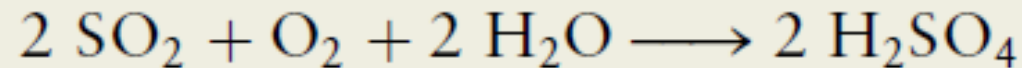
The one that is used up first: **limiting reactant**. These other reactants are present in **excess**.



limiting reactant

the silver nitrate is far more expensive than the sodium chloride

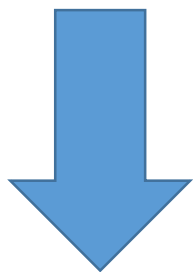
Sulfuric acid (H_2SO_4) forms in the chemical reaction



Suppose 400 g SO_2 , 175 g O_2 , and 125 g H_2O are mixed and the reaction proceeds until one of the reactants is used up. Which is the limiting reactant? What mass of H_2SO_4 is produced, and what masses of the other reactants remain?

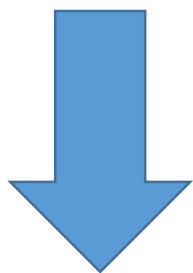
Percentage yield (收率)

Theoretical yields (理论产量) : the amounts of products calculated by assuming that the reaction goes cleanly and completely.



- Reaction stopped
- Competing reactions
- Lost during separation/purification

Actual yield (实际产量) < theoretical yield



The ratio of the actual yield to the theoretical yield (multiplied by 100%) gives the percentage yield for that product in the reaction.

The sulfide ore of zinc (ZnS) is reduced to elemental zinc by “roasting” it (heating it in air) to give ZnO, and then heating the ZnO with carbon monoxide. The two reactions can be written as



Suppose 5.32 kg ZnS is treated in this way and 3.30 kg pure Zn is obtained. Calculate the theoretical yield of zinc and its actual percentage yield.

Solution

From the molar mass of ZnS (97.46 g mol⁻¹), the number of moles of ZnS initially present is

$$\frac{5320 \text{ g ZnS}}{97.46 \text{ g mol}^{-1}} = 54.6 \text{ mol ZnS}$$

Because each mole of ZnS gives 1 mol of ZnO in the first chemical equation, and each mole of ZnO then gives 1 mol of Zn, the theoretical yield of zinc is 54.6 mol. In grams, this is

$$54.6 \text{ mol Zn} \times 65.39 \text{ g mol}^{-1} = 3570 \text{ g Zn}$$

The ratio of actual yield to theoretical yield, multiplied by 100%, gives the percentage yield of zinc:

$$\% \text{ yield} = \left(\frac{3.30 \text{ kg}}{3.57 \text{ kg}} \right) \times 100\% = 92.4\%$$

Homework: read “principle of modern chemistry”, chapter 2.